Rover Anomaly Review Board

Investigate a failed mission trial, identify the most likely cause, and approve a corrective action.

Advanced	1-2 class periods	Aerospace Mission

Engineering Context

During a rover mission rehearsal, the vehicle stopped early, missed the sample, or returned with incomplete data. Your team will act as an anomaly review board.

What You Will Practice

- Separate facts from assumptions
- Build a fault tree or cause map
- Use evidence to rank causes
- Define corrective and preventive actions

What You Need

- Instructor-provided anomaly scenario and evidence packet
- Fault-tree template
- Test logs or code excerpts
- Engineering notebook

Student Instructions

1. Create a timeline of confirmed events.
2. List facts, unknowns, and assumptions in separate columns.
3. Define the top failure event.
4. Build a fault tree or cause-and-effect diagram with at least three branches.
5. Use available evidence to eliminate or rank causes.
6. Identify the most likely root cause and one contributing factor.
7. Write a corrective action and a verification test.
8. Present a return-to-test recommendation: approve, approve with limits, or hold.

Engineering Requirements

- Use only evidence in the packet unless assumptions are labeled
- At least three plausible cause branches
- Corrective action must address the identified cause
- Verification test must have measurable pass/fail criteria
- Final recommendation includes residual risk

Constraints & Safety

- Do not blame individuals; focus on system, process, design, and evidence
- Protect any provided student/team identifiers
- Do not modify original logs

What You Submit

- Event timeline
- Facts/unknowns/assumptions table
- Fault tree
- Root-cause statement
- Corrective-action plan
- Return-to-test decision

Reflection Questions

1. What evidence most strongly changed your conclusion?
2. What cause remained possible but unproven?
3. How could the process prevent recurrence?
4. What additional data should future tests record?

Engineering Tip Preserve multiple plausible causes until evidence supports eliminating them.	**Common Mistake** Choosing the first obvious cause can lead to corrective action that does not prevent recurrence.
Stretch Goal Create a revised checklist or code monitor that detects the anomaly earlier.	**Career Connection** Aerospace anomaly review boards investigate test and mission failures before hardware returns to operation.